

Mohammad Ibrahim Saleem

+1(713) 853 7974 | Houston, TX | Ibrahimsaleem244@gmail.com | ibrahimsaleem.com | LD:ibrahimsaleem91 | Github:ibrahimsaleem

EDUCATION

University of Houston

Master of Science in Cybersecurity

Houston, TX

Expected August 2026

- GPA: 4.0/4.0; (Awarded 16k USD in Scholarship) Coursework: Network Security, Secure Enterprise Computing, Cryptography, Data Analysis for Cybersecurity, Cybersecurity Risk Management, Secure Software Design

Rajiv Gandhi Proudlyogiki Vishwavidyalaya

Bachelor of Technology in Computer Science Engineering

July 2023

- Coursework: Python, Data Structures & Algorithms, DBMS, Cloud Computing, Operating Systems, Machine Learning

EXPERIENCE

GenAI & Data Science Intern

June 2025 – Present

NOV Inc (National Oilwell Varco)

Houston, TX (On-site)

- Orchestrated end-to-end GenAI automation pipelines, reducing multi-day operational workflows to under 5 minutes, resulting in a 90% efficiency gain across internal teams.
- Developed and deployed an intelligent email responder integrated with an autonomous MCP server, enabling AI agents to self-heal and adapt, cutting response times by 75%.
- Enhanced large-scale data ingestion, validation, and preprocessing pipelines using Python and Pandas, improving AI-driven report accuracy by 20% and reducing downstream errors in analytics.

Research Assistant – AI & Cybersecurity

Sep 2024 – May 2025

University of Houston

Houston, TX (On-site)

- Designed and implemented **LIMA**, a modular LLM-powered penetration testing framework, achieving autonomous initial access on HackTheBox labs at \$0.05 per run, surpassing manual penetration testing efficiency.
- Performed advanced data analysis on large-scale IoT datasets using Python and MATLAB, applying signal processing and feature engineering to extract actionable insights, improving predictive model reliability by 30%.
- Conducted statistical analysis and experiment design for AI-driven security models, optimizing detection accuracy and system robustness.

Associate Software Engineer – Backend & Analytics

March 2023 – February 2024

Nagarro Software Pvt. Ltd.

- Developed secure backend systems using C#, .NET Core, and SQL Server, optimizing query performance by 30% and implementing JWT + RBAC authentication for fine-grained access control.
- Built interactive Angular dashboards and automated Python-based data pipelines, reducing manual reporting by 40% and enabling timely, data-driven insights for stakeholders.
- Collaborated with cross-functional teams to integrate analytics outputs into client-facing applications, improving operational efficiency and visibility across projects.

Data Science Intern – Cybersecurity Analytics

July 2022 – December 2022

State Cyber Cell MP Police

- Performed exploratory data analysis, statistical modeling, and anomaly detection on cybersecurity datasets to identify threats, reducing potential attack vectors by 25%.
- Developed interactive reports and visualizations using Python, improving operational insights by 35% and supporting evidence-based decision-making across security operations.
- Collaborated with law enforcement teams to translate complex analytical findings into actionable cybersecurity recommendations.

SKILLS & CERTIFICATIONS

- **Programming & Scripting:** Python (NumPy, Pandas, Matplotlib), R, SQL, MATLAB, JavaScript/TypeScript, Bash, PowerShell; proficient in writing efficient scripts for data processing, automation, and analysis.
- **Data Analysis & Visualization:** Advanced data wrangling, cleaning, and transformation using Pandas/NumPy; interactive and static visualizations with Matplotlib, Seaborn, Plotly; business intelligence tools such as Tableau and Power BI for dashboards and reporting.
- **Machine Learning & AI:** Supervised and unsupervised learning, feature engineering, model evaluation and hyperparameter tuning using scikit-learn, TensorFlow, PyTorch; Gradient Boosting frameworks (XGBoost, LightGBM); LLM fine-tuning (GPT, Llama), RAG pipelines, NLP applications, and Generative AI modeling.
- **Databases & Big Data:** Design and optimization of relational databases (MySQL, PostgreSQL, SQL Server) and NoSQL solutions (MongoDB, ChromaDB); experience with ETL/ELT pipelines, data warehousing, and managing large-scale datasets for analytics and ML workflows.
- **Cloud, DevOps & MLOps:** AWS (S3, Lambda, SageMaker, IAM), Microsoft Azure; Docker containerization, GitHub Actions, CI/CD pipelines for ML deployment; experience with model serving, monitoring, and version control.
- **Analytics, Statistics & Research:** Exploratory data analysis (EDA), statistical modeling, hypothesis testing, A/B testing, predictive analytics, and NLP; experience in designing experiments and deriving actionable business and security insights.
- **Certifications:** Google Data Analytics Professional Certificate, Microsoft Azure Data Fundamentals, Python for Data Science, Machine Learning Specialization, AI/ML MLOps courses from leading platforms, demonstrating continuous learning in emerging technologies.

PROJECTS

CareerPath AI – Personalized Learning Roadmaps | *Flask, GenAI, Python* | *CareerPath AI*

- Deployed an AI-driven web application generating personalized learning roadmaps for Cybersecurity, AI, and Data Science, delivering dynamic, context-aware explanations and improving user engagement by 45%.
- Integrated Flask backend with GenAI APIs to provide real-time, personalized recommendations and interactive visualizations, increasing retention and learning efficiency.

LocalRAGAgent – Offline Retrieval-Augmented QA | *Python, LangChain, FastAPI, Chroma, Ollama* | *LocalRAGAgent*

- Engineered a fully local RAG pipeline extracting text from PDFs, storing embeddings in ChromaDB, and serving responses via Llama-3 with 300ms latency, enhancing offline knowledge retrieval.
- Implemented secure, self-contained pipelines for sensitive enterprise data, enabling rapid, private access to contextual information without cloud dependency.

PentestThinkingMCP – Autonomous Pen-Testing Server | *MCP, Beam Search, MCTS, AI Agent* | *PentestThinkingMCP*

- Implemented an MCP server enabling LLM agents to autonomously plan and execute penetration tests, fully compromising HackTheBox Lame in 3 minutes at \$0.03 per run, demonstrating cost-efficient AI reasoning.
- Orchestrated multi-step attack simulations and vulnerability exploitation, validating the effectiveness of AI-driven cybersecurity frameworks.

Predictive Maintenance for IoT Devices | *Python, scikit-learn, Pandas, IoT*

- Built ML models predicting failure events in automotive IoT devices using historical sensor data, achieving 92% prediction accuracy.
- Designed a scalable data pipeline for real-time anomaly detection, reducing maintenance downtime by 25% and improving operational reliability.

WhatNext? – Movie Recommendation Web App | *React, Flask, ML, Firebase* | *WhatNext*

- Developed a recommendation engine using cosine similarity and ML models, delivering personalized suggestions based on user preferences and viewing history.
- Deployed backend APIs on Flask and frontend on Firebase, ensuring smooth, low-latency interactions and responsive UI across devices.

Network Traffic Anomaly Detection | *Python, TensorFlow, scikit-learn, Wireshark*

- Built ML-based classifiers to detect abnormal network traffic and potential security threats, achieving over 95% accuracy in real-world simulations.
- Integrated visualization dashboards to monitor live traffic patterns and automate alerts for suspicious activity, improving incident response efficiency by 30%.

Cybersecurity Awareness Platform | *Python, JavaScript, Flask* | *Cybersecurity Awareness Platform*

- Launched an interactive training platform with 30+ hands-on labs, improving learner engagement and cybersecurity knowledge retention by 40%.
- Developed simulations of real-world cyber threats and attack scenarios, providing practical, gamified training for students and professionals.

ApplyEase – Job Application Automation Platform | *Flask, Python, GenAI* | *ApplyEase*

- Built an AI-powered platform to automate job applications, generating customized resumes and cover letters based on user profiles and job descriptions.
- Integrated automated submission workflows across multiple portals, increasing application efficiency and success rates by 50%.

RESEARCH

- **Saleem, M. I.** (First Author), Prabhakar, S. S., Harsha, A., Nagabhushan, D., Conklin, W. A., Lee, K. I., Banerjee, T. “*LIMA: Leveraging Large Language Models and MCP Servers for Initial Machine Access.*” *Proc. IEEE Intl. Conf. on Future Machine Learning and Data Science (FMLDS)*, 2025 (Accepted)
- NOV GenAI Team, **Saleem, M. I.** (Second Author). “*Self-Improving GenAI Agent for Fully Automated Report Parsing in Enterprise Environments.*” Research proposal submitted to the **Society of Petroleum Engineers (SPE)**, 2025 — conducted as *GenAI & Data Science Intern, NOV Inc.* (Accepted)
- PHM Society Data Challenge 2025 — Collaborated with NOV GenAI team to develop ML models predicting aircraft engine maintenance events. Submission planned for PHM North America 2025 Conference.